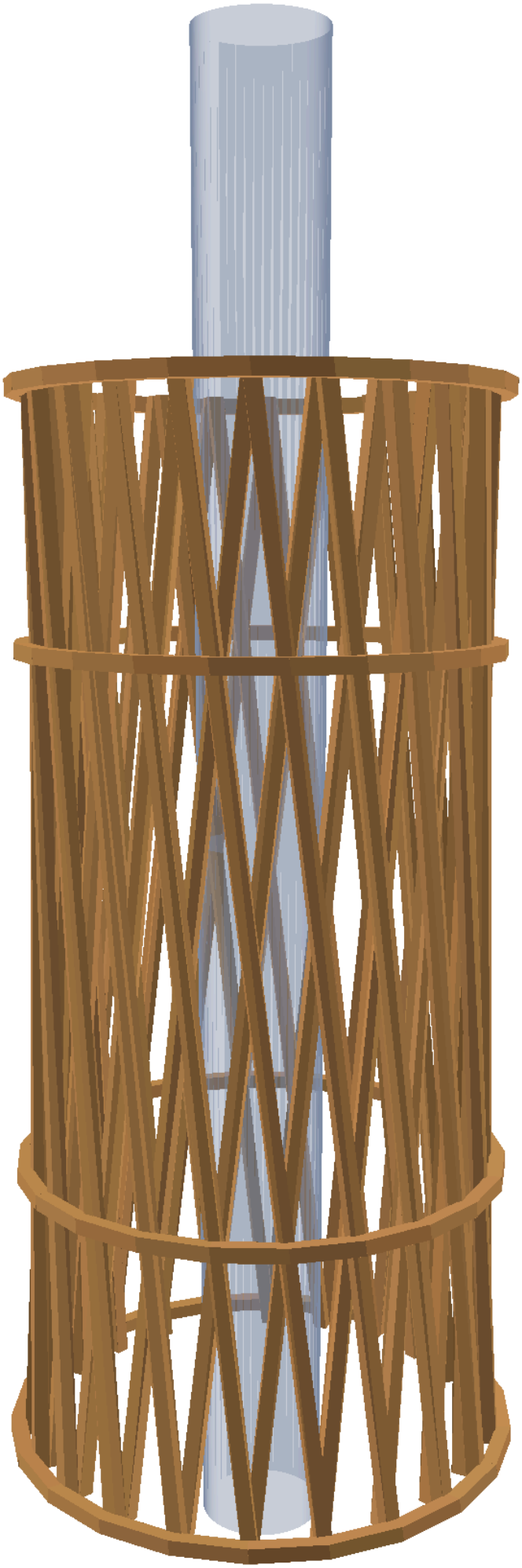
3×3 twisted diagrid

Assembly guide

Generated by Tower Designer. Brief: ≤ 20 g, 211 mm tall, fits over a 25 mm pole, built only from supplied 3 × 3 mm sticks (full size; only the joints are shaped).



Predicted failure load

60 kg

Predicted mass

19.03 g

wood 15.62 g + glue 3.41 g, including +3 % on every piece and joint

132 pieces · 286 glue joints · 13 sticks

(assuming 900 mm sticks)

Weakest point: Strip A11 buckles.
Gap to the pole: 24.0 mm all round.

Strength is based on **book values for medium balsa** because the stick species is unknown. Do the wood test first (section 5).

Contents

1. How to use this guide
2. How the tower works (2b: how the rings attach)
3. Safety
4. What you need
5. Before you start: test your wood and glue
6. Wood and grain direction
7. Glue plan
8. Printing the jig
9. Cutting the pieces
10. Build steps
11. Final checks
12. Test day
13. Troubleshooting and fixing mistakes
14. Appendix A: every piece
15. Appendix B: jig and model files

1. How to use this guide

- Every piece has an ID: A1–A22 and B1–B22 for the strips, R1-1, R3-1 … for the rings (ring number, then piece), BASE…, CAP… . Write it on the piece in pencil when you cut it.
- Every build step has a picture **and a 3D model** in the `steps/` folder. Open the models in Onshape (**+** → **Import**) and orbit round them if a picture isn't clear. In every model the **jig is blue**, pieces you have **already built are tan**, and **the pieces you add in that step are orange**.
- Every step lists the **jig parts** it uses by their engraved ID (dark tag, e.g. **SP2**) and file name in `jig/`.
- **Glue colour code**, the same in every picture, model, table and step: each glue point is a coloured mark.

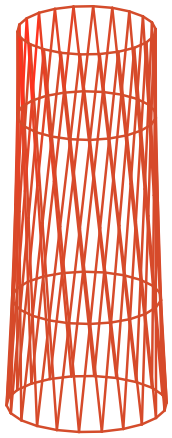
> Glue dots: 🔵 **PVA wood glue** 🔴 **Balsa cement**

- ⚖ boxes are weigh-ins. Weigh the tower after each step once its glue is dry. If you are over, you are using too much glue.
- ☐ boxes are checks to tick before you move on. ⏸ boxes mean wait for the glue.

Words used in this guide

Strip	One of the long sticks that run from the table to the top, leaning (and twisting slightly).
Set A / set B	The two sets of strips. A strips are the inner layer and lean one way; B strips are the outer layer and lean the other way. The pieces are identical until you letter them.
Letter face	The face you write A or B on. It always faces outwards on the tower.
Foot (F)	The end of a strip that stands on the table. You mark it F.
Crossing	Where a B strip passes over an A strip; they are glued face to face there.
Level	One of the heights where the strips cross (1 = lowest). An R after the number means a ring goes there.
Tick	A pencil line on a strip, made with the marking gauge MG1, showing where a crossing goes.
Ring	A hoop of short pieces glued round the outside at one level.
Level cut	An angled end cut so a leaning strip stands flat on a level surface.
Lap joint	Two pieces glued face to face where one overlaps the other.
Pocket	A shallow V-shaped hole in the base plate where two strip feet stand.
Collar / segment	A printed ring that holds the strips at the right place; it comes in numbered segments so it can be taken out afterwards.
Notch	The slot in a collar segment that a strip presses into.
Mast / spacer	The rod up the middle of the jig, and the tubes stacked on it that set each collar's height. Each spacer has a disc on top that the collar segments rest on.
Grain	The direction of the wood fibres. In a stick it runs along its length.
End grain	The cut end of a stick. It soaks up glue and bonds poorly.
Double-glue	Balsa cement method: coat both faces, let it go tacky, coat again and press.
Cure	The time a glue needs to reach full strength.
Buckling	A slender piece bowing sideways under compression; the usual way thin sticks fail.

2. How the tower works



Colour = how hard each piece works at the predicted failure load (green light → red at its limit).

22 strips twist one way (**set A**, the inner layer) and 22 twist the other way (**set B**, the outer layer). Each strip turns through 33° on its way up. The base is Ø91 mm and the top Ø78 mm, measured to the strip centres, so the tower narrows like a cooling tower. Feet are 16.4° apart round the base. The strips cross at 3 levels: 60, 113, 163 mm above the table. Rings sit at some of those levels. The printed jig sets every one of these positions for you.

Load path. The load plate presses on the top ring and the strip tops. Each strip carries its share straight down to the table. The strips lean in opposite directions, so where an A strip crosses a B strip they brace each other. That shortens the length that can buckle and turns the whole wall into a stiff lattice shell, like a woven basket.

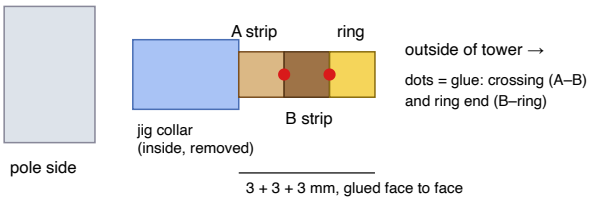
Why it narrows. A wider base resists tipping and buckling of the whole tower. The narrower top keeps the strips short and steep, so they stay in compression.

What the rings do. They hold the crossings at their levels so the lattice keeps its shape, and stop the strips spreading outwards at the base and squeezing inwards at the top.

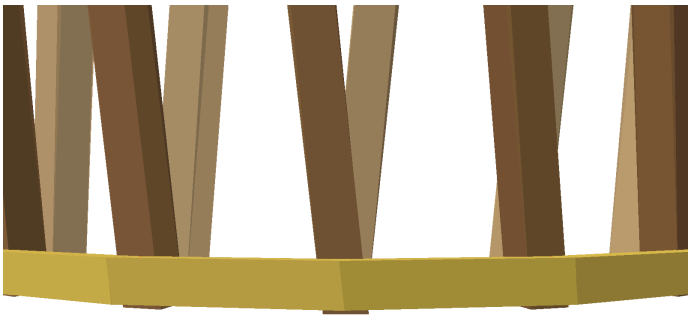
Why lap joints. Every joint is one face glued flat on another (side grain to side grain). That is far stronger than gluing onto a cut end.

How it is predicted to fail. Strip A11 buckles. The whole tower would buckle at about 76 kg, well above the failure load.

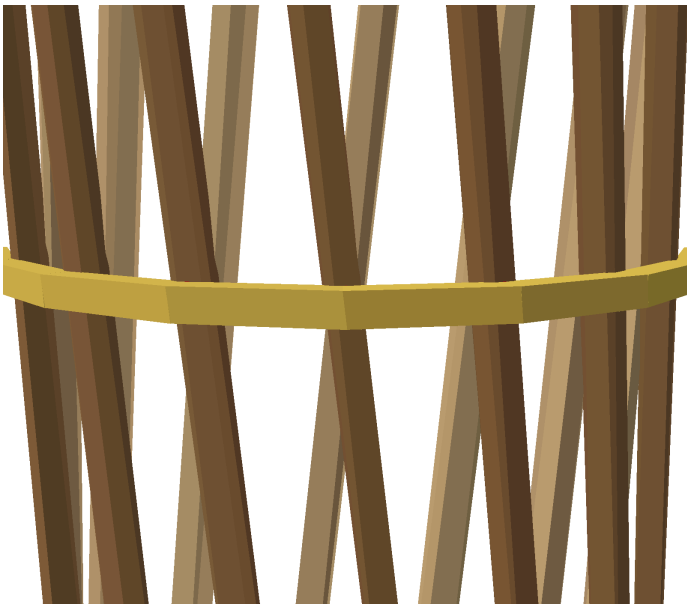
2b. How the rings attach, and why nothing hits the jig



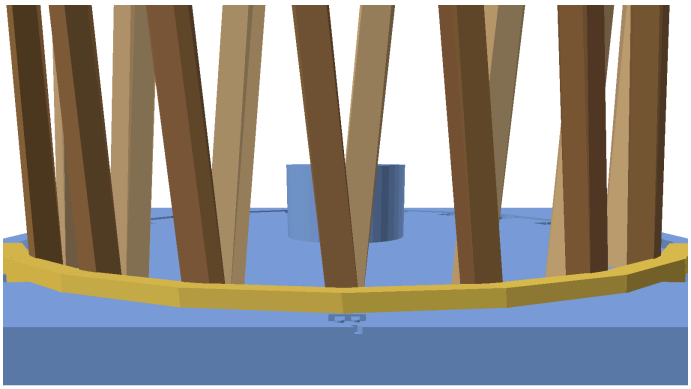
The tower is built in **three layers**, from the inside out: the A strips, then the B strips glued over them at every crossing, then the rings glued on the outside of the B strips. Each layer is one 3 mm stick thick, and every joint is a face glued flat on a face.



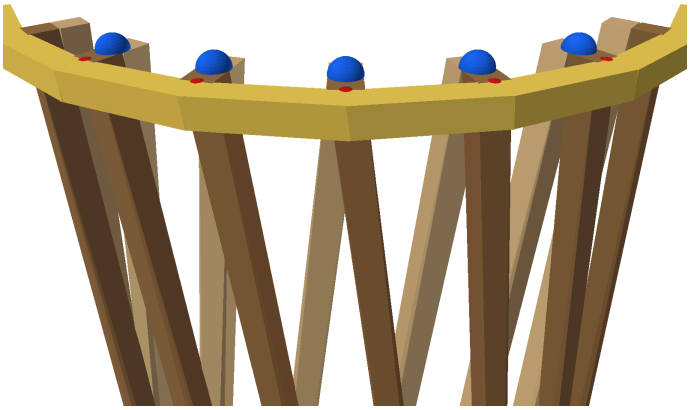
A crossing. The B strip lies over the A strip; glue goes where they touch (dot).



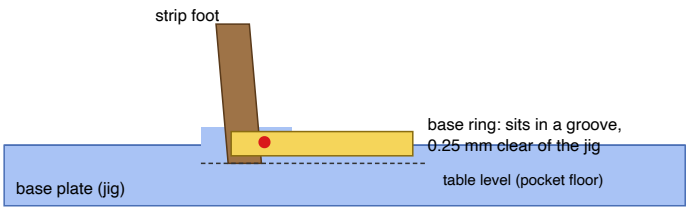
A ring at a crossing level. Ring pieces (gold) lie on the B strips. Two pieces meet end to end on each B strip, each covering half its width, and each end is glued (dots).



The base ring in the jig. It lies on the outside of the feet, in a shallow groove in the base plate (blue) that keeps it 0.25 mm clear of the plastic.



The top ring. Its top is flush with the strip tops so the load plate presses on both.



Why the base ring doesn't hit the jig:

- The strip feet stand in 2 mm-deep pockets in the base plate. The base ring sits just above the feet on the outside, in its own groove, with 0.25 mm of air all round, so it is never glued to or trapped by the plate.
- The collars hold the strips from the **inside**; every ring is on the **outside**. The lowest collar starts 27 mm up, well above the base ring.
- Collars sit between the ring levels, never at the same height.
- Checked in the 3D model: no jig part overlaps any piece of the tower anywhere.
- When you lift the finished tower off, the feet come straight up out of their pockets and the base ring out of its groove.

3. Safety

- Wear safety glasses when cutting, sanding, and crush-testing (wood splinters).
- Glue: work with a window open. Balsa cement gives off solvent fumes and is flammable: keep it away from flames and put the lid back on between joints. Acetone (nail-polish remover, for fixing mistakes) is flammable too.
- Razor saws and hobby knives: cut away from your fingers, on a cutting mat, with the stick held in the guide.
- Let the 3D printer cool before removing prints.

4. What you need

Materials

- **15 × 3 × 3 mm sticks**: 13 plus 2 spare (stick plan in section 9).
- Selleys Aquadhere Quick Set PVA (interior) — Bunnings
- UHU Hart balsa cement — hobby / model shops (not Bunnings)
- PLA filament for the jig, about 150 g.
- Packing tape, masking tape, baking paper.
- Acetone (nail-polish remover) and cotton buds, for undoing a wrong joint.

Tools

- Fine razor saw (or a new hobby-knife blade) and a cutting mat.
- 240-grit sandpaper glued flat to a board, plus a small sanding stick.
- Scale reading to 0.01 g. Ruler. Calipers if you have them.
- A small brush or toothpicks, small cups (one per glue), scrap card, small spring pegs, sharp pencil, fine marker.
- 6 small cups or bags for sorting the pieces.
- Bambu printer (jig) and the app on the Mac.

5. Before you start: test your wood and glue

The prediction above uses book values and a standard glue allowance. Two quick tests make it match **your** sticks and **your** gluing. Both kits are in the `test kits/` folder of this build pack.

Wood test (`Wood test kit/`, about 30 min)

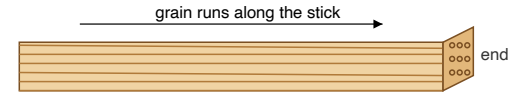
1. Weigh 5 whole sticks: this gives the density.
2. Crush 5 pieces 12 mm long, ends sanded square in `wood_test_squaring_block.stl`, between `wood_test_crush_pad.stl` pads on a 25 kg scale. Record each peak.
3. Press 3 straight 150 mm pieces in `wood_test_buckling_cap.stl` caps until they bow, and record the load.
4. Enter the results in the app (Wood and glue → Wood test…) and click Apply.

Glue test (`Glue test kit/`, a day for curing)

1. For each glue, weigh a 100 mm rod and 5 short strips together.
2. Glue the strips onto the rod in `glue_test_jig_base.stl` + `glue_test_jig_comb.stl`, exactly as you'll glue the tower.
3. When dry, weigh again. Enter the masses in the app (Glue test…) and click Apply.

After either test, **re-run the search and export the build pack again**. The design, cut list, and this guide may change.

6. Wood and grain direction



Side grain: long faces (good glue surface, strong)
End grain: the cut end (soaks up glue, weak bond)

The grain (the wood fibres) runs along the length of every 3 × 3 stick. Wood is strong and stiff along the grain: about **20–30× stiffer** than across it, and that is the direction every piece of this tower carries its load. **Never split or rip a stick lengthwise.** Only cut across it, so the grain always runs along each piece.



✔ grain parallel to the edges

✘ grain runs off the edge (snaps)

Pick straight-grained sticks. Hold each stick to the light and look at all four faces. Grain lines should run parallel to the edges. If they run off an edge ("run-out"), the stick will snap there. Use those sticks for short ring pieces only, cutting round the run-out.

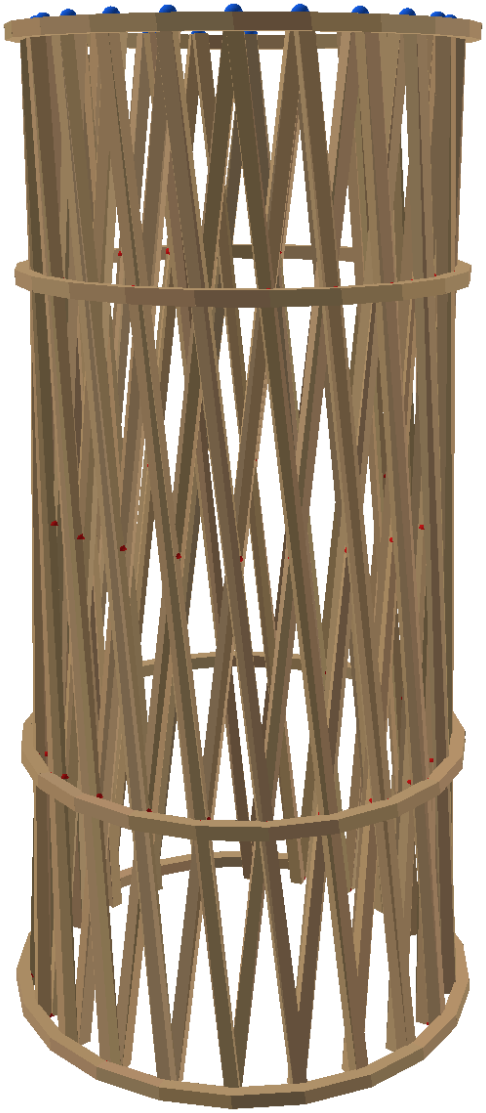
Sort the sticks before cutting:

1. Weigh every stick and write its mass on it. Mass ÷ volume (length × 9 mm²) gives the density; book balsa is about 160 kg/m³.
2. Rank them heaviest to lightest. Heavier (denser) sticks are stiffer and stronger.
3. Use the **heaviest 11 sticks** (sticks 1–11 in the stick plan) for the strips. They carry the load.
4. Use the lightest sticks for ring pieces, which carry little load.
5. Reject any stick that is bent, cracked, or much lighter than the others.

At the joints: glue side grain to side grain. A cut end (end grain) soaks up glue and bonds at only about 60 % of the strength, which is why this design laps every joint face-on-face. Keep sticks dry and flat; don't leave them in the sun or a damp room before building.

7. Glue plan

Glue dots: ● **PVA wood glue** ● **Balsa cement**



Glue map: every glue point on the tower, coloured by the glue to use. Each step's picture and 3D model shows the same dots.

Each joint gets the lightest glue that is still strong enough for the load it carries. "Apply" is roughly what to put on; "dries to" is the average most that should stay in each joint (joints with a bigger face get a little more). A pinhead dab is about 2–4 mg.

Where	Joint	Glue	Count	Apply (wet, average)	Dries to (average)
Base ring → the strips (lapped across the outside)	lap end	Balsa cement	44	≈ 12 mg	≤ 4 mg
Crossings (B strip over A strip)	lap	Balsa cement	88	≈ 74 mg	≤ 22 mg
Crossings (B strip over A strip)	lap	PVA wood glue	22	≈ 73 mg	≤ 37 mg
Ring 1 → the strips (lapped across the outside)	lap end	Balsa cement	44	≈ 12 mg	≤ 4 mg
Ring 3 → the strips (lapped across the outside)	lap end	Balsa cement	44	≈ 12 mg	≤ 4 mg
Top cap → the strips (lapped across the outside)	lap end	Balsa cement	44	≈ 12 mg	≤ 4 mg

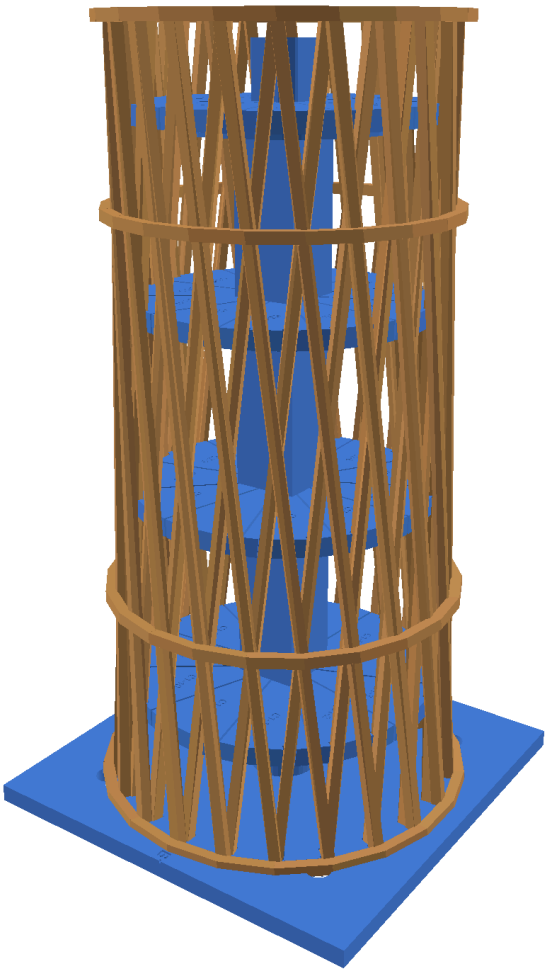
How to use each glue

Glue	Method	Cure	In this tower
PVA wood glue Selleys Aquadhere Quick Set PVA (interior) — Bunnings	Thin coat on one face only, press together for 30–60 s, then hold with a rubber band or peg for 20 min. Full strength after 24 h.	Hold 10–20 min, full strength after 24 h	22 joints, 0.81 g dry
Balsa cement UHU Hart balsa cement — hobby / model shops (not Bunnings)	Double-glue. Brush a thin coat on both faces, wait 5–10 min until it is tacky-dry (it soaks into the wood), add a second thin coat to one face and press together for 30 s. Leave 12 h before loading.	Tacky in 1–2 min, full strength after a few hours	264 joints, 2.60 g dry

Preparing a joint: make sure the two faces touch flat with no light showing between them. If they don't, touch the contact area (only) once on the sandpaper. Blow off dust before gluing. Don't fill gaps with extra glue; fix the fit instead.

If you'd rather not use balsa cement: with PVA on every joint, this tower would weigh 24.42 g and hold 60 kg — over the limit. To switch, set Glue to PVA only in the app, re-run the search, and export again.

8. Printing the jig



All 55 files are in `jig/` as **STL** (for Bambu Studio) and **STEP** (for Onshape). `jig/jig_assembly.step` shows the whole jig assembled.

Print settings (PLA): 0.2 mm layers, 3 walls, 15 % infill, no supports, brim off. Print each part in the orientation in the table below: collar segments with the engraved ID facing up (the notches lean, so upside down is wrong), spacers disc-down, the mast lying flat.

Print in this order (rough times on a Bambu A1/P1):

Plate	What	Parts	About
1	Test first	FT	≈ 5 g, 0.4 h
2	Preparation jigs	CG1, MG1, SJ1, SJ2	≈ 123 g, 11.2 h
3	Base plate	BP	≈ 35 g, 3.1 h
4	Mast and spacers	M, SP1, SP2, SP3, SP4	≈ 43 g, 3.9 h
5	Collar C1	C1-1, C1-2, C1-3, C1-4, C1-5, C1-6, C1-7, C1-8, C1-9, C1-10, C1-11	≈ 15 g, 1.4 h
6	Collar C2	C2-1, C2-2, C2-3, C2-4, C2-5, C2-6, C2-7, C2-8, C2-9, C2-10, C2-11	≈ 13 g, 1.2 h
7	Collar C3	C3-1, C3-2, C3-3, C3-4, C3-5, C3-6, C3-7, C3-8, C3-9, C3-10, C3-11	≈ 12 g, 1.1 h
8	Collar C4	C4-1, C4-2, C4-3, C4-4, C4-5, C4-6, C4-7, C4-8, C4-9, C4-10, C4-11	≈ 12 g, 1.1 h

After printing:

1. Print **FT** first and check your sticks fit (Step 0, item 1) before printing anything else.
2. Remove any stringing from the notches and slots with a knife.
3. Check every part's engraved ID is readable (C1-1, C1-2 …, SP1 …); go over faint ones with a fine marker.
4. Stack the jig dry on the mast (Step 1) once before you start, to check every segment sits on its number.

Every part has its **ID engraved** on it (the bold tag below). The guide uses these IDs everywhere.

ID	File	Qty	Used in	Print orientation
BP	`base_plate.stl`	1	steps 1–7	flat face down
C1-1 … -11	`collar_1_seg1 … seg11.stl`	11	steps 1–7	engraved ID facing up
C2-1 … -11	`collar_2_seg1 … seg11.stl`	11	steps 1–7	engraved ID facing up
C3-1 … -11	`collar_3_seg1 … seg11.stl`	11	steps 1–7	engraved ID facing up
C4-1 … -11	`collar_4_seg1 … seg11.stl`	11	steps 1–7	engraved ID facing up
CG1	`cutting_guide_1.stl`	1	Step 0 (preparing parts)	flat face down
FT	`fit_test_coupon.stl`	1	Step 0 (preparing parts)	flat face down
M	`mast.stl`	1	steps 1–7	lying flat on the bed
MG1	`marking_gauge.stl`	1	Step 0 (preparing parts)	flat face down
SJ1	`end_sanding_jig_7deg.stl`	1	Step 0 (preparing parts)	wide flat bottom down
SJ2	`end_sanding_jig_rings.stl`	1	Step 0 (preparing parts)	wide flat bottom down
SP1	`spacer_1.stl`	1	steps 1–7	disc down (upside down)
SP2	`spacer_2.stl`	1	steps 1–7	disc down (upside down)
SP3	`spacer_3.stl`	1	steps 1–7	disc down (upside down)
SP4	`spacer_4.stl`	1	steps 1–7	disc down (upside down)

Stack on the mast, bottom to top (heights above the table):

Part	Sits at
BP base_plate.stl boss	up to 14.0 mm
SP1 spacer_1.stl (13.5 mm tall, disc on top)	14.0 → 27.5 mm
C1-1 … C1-11 collar_1_seg1 … seg11.stl, each on its number	27.5 → 32.5 mm
SP2 spacer_2.stl (51.6 mm tall, disc on top)	32.5 → 84.1 mm

C2-1 … C2-11 collar_2_seg1 … seg11.stl, each on its number	84.1 → 89.1 mm
SP3 spacer_3.stl (46.5 mm tall, disc on top)	89.1 → 135.6 mm
C3-1 … C3-11 collar_3_seg1 … seg11.stl, each on its number	135.6 → 140.6 mm
SP4 spacer_4.stl (43.8 mm tall, disc on top)	140.6 → 184.4 mm
C4-1 … C4-11 collar_4_seg1 … seg11.stl, each on its number	184.4 → 189.4 mm

9. Cutting the pieces

Every piece is a full 3 × 3 stick cut to length; only the ends are shaped. The cutting guide **CG1** has one channel per piece, numbered and labelled with its length. Each channel is open at the left, has a saw slot near the left end and a **stop** exactly one piece-length to the right of the slot. You never measure.

1. Tape the guide to the bench. Pick the channel for the piece (table below).
2. **Square up the start.** Lay the stick in the channel from the left so its end pokes about 5 mm past the slot, and saw through the slot. Throw the offcut away.
3. **Cut a piece.** Slide the stick to the right until its fresh end touches the stop. Hold it there with a finger well away from the slot, and saw gently through the slot. The piece between the slot and the stop is exactly the right length, and the end left on the stick is already cut for the next piece.
4. Repeat step 3 along the stick. **Don't turn the stick over between cuts**, so the angled strip ends stay parallel.
5. **Channels with two angled slots** (pieces whose two ends are cut at different angles, like the ring pieces): push the stick against the stop at the left end of the channel and saw at **both** slots, keeping the saw against the **waste side** of each slot (the side away from the piece): the slots are spaced so the piece comes out exactly to length after the saw cut. Before you take the piece out, pencil an **arrow on its top face pointing to the right-hand end**. In the ring, that face goes outwards with the arrow pointing right; the two ends are cut at different angles, so this is the only way round it fits.
6. Write the piece ID on it in pencil and drop it in its cup. Strips also get their letter and F (Step 0).

Piece	Qty	Length	Tolerance (mm)	End 1	End 2	Cut with	Sand ends with
Strip A A1…A22	22	**212.4 mm**	cut 212.7, sand to 212.4 ±0.3	level cut, ~7° off square	level cut, ~7° off square	**CG1** ch 1	**SJ1**
Strip B B1…B22	22	**212.4 mm**	cut 212.7, sand to 212.4 ±0.3	level cut, ~7° off square	level cut, ~7° off square	**CG1** ch 1	**SJ1**
Base ring BASE1…BASE22	22	**14.6 mm**	±0.3 (end gap ≤ 0.6)	lap end, trimmed 5° off square	lap end, trimmed 6° off square	**CG1** ch 2	**SJ2**
Ring 1 R1-1…R1-22	22	**13.7 mm**	±0.3 (end gap ≤ 0.6)	lap end, trimmed 6° off square	lap end, trimmed 6° off square	**CG1** ch 3	**SJ2**
Ring 3 R3-1…R3-22	22	**12.9 mm**	±0.3 (end gap ≤ 0.6)	lap end, trimmed 6° off square	lap end, trimmed 7° off square	**CG1** ch 4	**SJ2**
Top cap CAP1…CAP22	22	**12.8 mm**	±0.3 (end gap ≤ 0.6)	lap end, trimmed 7° off square	lap end, trimmed 7° off square	**CG1** ch 5	**SJ2**

Tolerances: how accurate each cut has to be

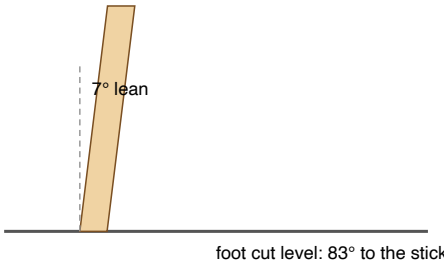
Every size in this guide has a tolerance: how far off it can be and still fit. The jigs are made with these allowances built in, so if you use the jig as described you will be inside them.

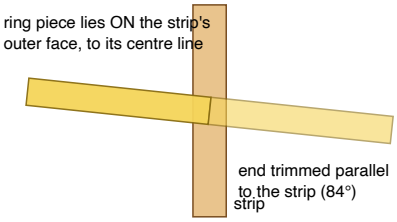
What	Nominal	Tolerance	How the jig allows for it	How to check
Stick size	3.0 × 3.0 mm	**±0.15 mm (typical)**	Every jig slot is 3.3 mm (0.15 mm clear each side). Set it from the fit-test coupon FT, which also cancels out your printer's error.	FT: the stick slides in with light friction.
Printed jig slots and holes	as designed	**±0.15 mm**	Collar notches and base pockets have 0.25 mm clearance each side; the mast hole 0.2 mm each side; the mast socket 0.1 mm each side.	Step 1 dry stack: every strip drops into its notch without forcing.
Strip length (cut)	212.7 mm	**±0.3 mm**	The cutting-guide stop is set 0.3 mm long so the sanding jig can clean both ends.	No measuring: the stop sets it.
Strip length (after sanding)	212.4 mm	**±0.3 mm**	Marking gauge MG1 has an END notch with a small tick 0.3 mm either side.	The strip's top end lands between the two small ticks.
Strip end angle	level cut, 6.6° off square in total (lean + inward tilt)	**±0.8° (≤ 0.04 mm across the end)**	Sanding jig SJ1 holds the strip at the exact compound angle; the slot clearance sets the tolerance.	The strip stands on the table at its lean without rocking.
Strip twist	33° from foot to top	**±3°**	Every collar notch is angled to the twist at its height, with 0.25 mm clearance; damp strips take the twist without strain.	Each strip lies flat in every notch, no edge lifting.
Crossing gap before pegging	0 mm	**up to 0.4 mm**	A straight strip can't sit exactly on every curved crossing; the 3D check shows gaps of up to about 0.3 mm. A peg bends the strip that little to close it.	With the peg on, no light between the faces (gap under 0.1 mm).
Crossing marks	at each level	**±0.5 mm**	MG1 notches are 0.8 mm wide: put the pencil in the middle.	A and B ticks line up at every crossing in the dry fit.
Ring piece length	see cut list (strip centre to strip centre)	**±0.3 mm**	The cutting guide cuts both ends at their angle against a stop, so neighbouring pieces meet on the middle of each strip. A short piece leaves at most 0.6 mm between the two ends; each end is still glued over its half of the strip. A long piece won't sit: sand its end lightly in the ring-end sanding jig.	Both ends sit on the middle of their strips; the gap between neighbouring ends is no more than 0.6 mm.
Ring end angle	see cut list	**±1°**	The cutting-guide slot sets the angle; the ring-end sanding jig cleans it up.	Two neighbouring ends meet with no light between them.
Ring height	at a crossing level	**±0.5 mm**	The crossing (pencil tick) sets it; no measuring.	The ring looks level against the ticks all round.
Tower height	211.0 mm	**–0.5 / +0.3 mm**	Strips are cut to full height; the top is only lightly sanded flat. Worst case is 210.5 mm, still over the 210 mm minimum.	Measure at three places round the top.
Gap to the pole	24.0 mm all round	**–1.0 mm (if the tower is built leaning)**	The collars hold every strip at its radius.	The pole slides through without touching; at least 23.0 mm even with a lean.
Base ring in its groove	0.25 mm clear all round	**±0.15 mm (print)**	The groove is the ring's own shape grown by 0.25 mm, so the ring never touches the plastic; the strips hold it.	The ring isn't touching the base plate anywhere.

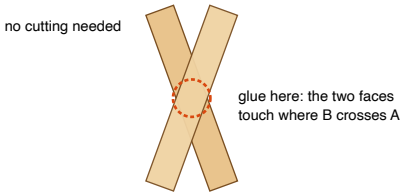
Fit check of the 3D model (run automatically when this pack was made):

Check	Result
✔ Crossings, feet and tops: B strip glued to A strip	110 joints: glue face touching 97 % on average, worst 92 % (within 0.2 mm, pressed); deepest overlap 0.00 mm; faces within 0.1° of parallel; largest gap before pegging 0.11 mm (the strip bends to close it)
✔ Ring ends glued to their strip	176 joints: glue face touching 100 % on average, worst 100 % (within 0.2 mm, pressed); deepest overlap 0.12 mm
✔ Neighbouring ring pieces meet on each strip	88 joints: largest gap 0.00 mm, largest overlap 0.00 mm³
✔ Jig never overlaps the tower	0 overlaps

How it is checked: every glued pair is measured in the 3D model, with the strips twisted to the surface as they really sit. A face counts as touching where it is within 0.2 mm once the joint is pegged or pressed (balsa squeezes that much). A gap before pegging means the straight strip bends that little when you peg it. A hairline overlap of a few tenths of a millimetre means the parts press together, which is what a glue joint wants.







- **Level cuts (strip ends):** the strips lean, so both ends are cut at the same angle and parallel to each other. The cutting guide gets the angle close. The sanding jig **SJ1** then makes it exact, including the slight inward tilt of the tower.
- **Ring ends:** Two pieces meet end to end on each B strip, each covering half its width, and each end is glued (dots). Trim the end parallel to the strip so the two pieces butt neatly.
- **Crossings:** no cutting at all. The faces touch where B crosses A; you only glue there.
- Cut 2 or 3 spares of each ring length. Ring pieces are tiny and easy to lose or break.

Stick plan (which pieces come from which stick, in mm, allowing 5 mm to square up the start, 2 mm per saw cut, and 3.5 mm for pieces cut between two angled slots). Cut the strips from the heaviest sticks:

- Stick 1: A1 (212.4), A2 (212.4), A3 (212.4), A4 (212.4), BASE16 (14.6), BASE20 (14.6)
- Stick 2: A5 (212.4), A6 (212.4), A7 (212.4), A8 (212.4), BASE10 (14.6), BASE13 (14.6)
- Stick 3: A9 (212.4), A10 (212.4), A11 (212.4), A12 (212.4), BASE3 (14.6), BASE12 (14.6)
- Stick 4: A13 (212.4), A14 (212.4), A15 (212.4), A16 (212.4), BASE8 (14.6), BASE22 (14.6)
- Stick 5: A17 (212.4), A18 (212.4), A19 (212.4), A20 (212.4), BASE2 (14.6), BASE4 (14.6)
- Stick 6: A21 (212.4), A22 (212.4), B1 (212.4), B2 (212.4), BASE1 (14.6), BASE7 (14.6)
- Stick 7: B3 (212.4), B4 (212.4), B5 (212.4), B6 (212.4), BASE5 (14.6), BASE18 (14.6)
- Stick 8: B7 (212.4), B8 (212.4), B9 (212.4), B10 (212.4), BASE14 (14.6), BASE21 (14.6)
- Stick 9: B11 (212.4), B12 (212.4), B13 (212.4), B14 (212.4), BASE6 (14.6), BASE17 (14.6)
- Stick 10: B15 (212.4), B16 (212.4), B17 (212.4), B18 (212.4), BASE9 (14.6), BASE19 (14.6)
- Stick 11: B19 (212.4), B20 (212.4), B21 (212.4), B22 (212.4), BASE11 (14.6), BASE15 (14.6)
- Stick 12: CAP1 (12.8), CAP5 (12.8), CAP8 (12.8), CAP10 (12.8), CAP12 (12.8), CAP13 (12.8), CAP16 (12.8), CAP20 (12.8), CAP22 (12.8), R1-1 (13.7), R1-2 (13.7), R1-3 (13.7), R1-4 (13.7), R1-5 (13.7), R1-6 (13.7), R1-7 (13.7), R1-8 (13.7), R1-9 (13.7), R1-10 (13.7), R1-11 (13.7), R1-12 (13.7), R1-13 (13.7), R1-14 (13.7), R1-15 (13.7), R1-16 (13.7), R1-17 (13.7), R1-18 (13.7), R1-19 (13.7), R1-20 (13.7), R1-21 (13.7), R1-22 (13.7), R3-1 (12.9), R3-2 (12.9), R3-3 (12.9), R3-4 (12.9), R3-5 (12.9), R3-6 (12.9), R3-7 (12.9), R3-8 (12.9), R3-9 (12.9), R3-10 (12.9), R3-11 (12.9), R3-12 (12.9), R3-13 (12.9), R3-14 (12.9), R3-15 (12.9), R3-16 (12.9), R3-17 (12.9), R3-18 (12.9), R3-19 (12.9), R3-20 (12.9), R3-21 (12.9), R3-22 (12.9)
- Stick 13: CAP2 (12.8), CAP3 (12.8), CAP4 (12.8), CAP6 (12.8), CAP7 (12.8), CAP9 (12.8), CAP11 (12.8), CAP14 (12.8), CAP15 (12.8), CAP17 (12.8), CAP18 (12.8), CAP19 (12.8), CAP21 (12.8)

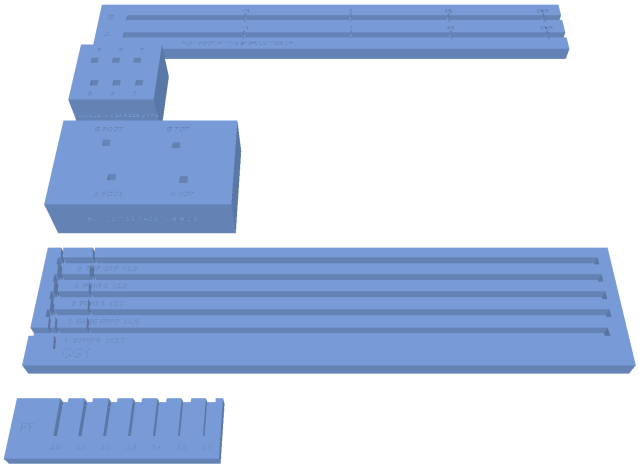
10. Build steps

Before Step 1: all pieces cut and labelled, jig printed and test-fitted, glues and tools laid out, a clear flat bench.

A sensible plan (each session 1–2 hours):

Session	Do	Then wait
1	Step 0: test fit, cut, letter, mark and sand every piece	—
2	Steps 1–2 (jig and A strips, no glue), then the first half of Step 3	1 h between batches
3	The rest of Step 3 (crossings)	12 h
4	Base ring and the first ring	1 h
5	The other rings and the top ring	24 h
6	Step 8: remove the jig, finish, weigh	—

Step 0. Preparing the parts



The preparation jigs, each with its ID engraved.

Jigs for this step:

- FT `fit_test_coupon.stl` : finds the slot size your sticks need. Print it first.
- CG1 `cutting_guide_1.stl` : one channel per piece, each with a saw slot and a stop one piece-length away. Channel numbers and lengths are engraved beside each channel.
- MG1 `marking_gauge.stl` : marks where every crossing and ring goes on a strip, and checks its length.
- SJ1 `end_sanding_jig_7deg.stl` : Strip ends, 6.6° off square in total (the lean and the inward tilt together). Four slots: A foot, A top, B foot, B top. Letter face to the front.
- SJ2 `end_sanding_jig_rings.stl` : Sands the angled ends of Base ring, Ring 1, Ring 3, Top cap. Each angle (degrees off square) has two slots, one leaning each way: use the one that matches the cut end. Push the piece down until it touches the sandpaper, hold it still and rub until the plastic scuffs.

Also: razor saw, 240-grit sandpaper glued flat to a board, sharp pencil, 0.01 g scale, 6 small cups or bags.

Everything you need to make (132 pieces from 13 sticks). Tick each row when its pieces are cut, sanded and labelled:

Piece	Qty	Length (tolerance, mm)	End 1	End 2	Cut with	Sand with	From stick	Group mass
□ **Strip A** A1…A22	22	**212.4 mm** cut 212.7, sand to 212.4 ±0.3	level cut, ~7° off square	level cut, ~7° off square	CG1 ch 1	SJ1	1–6	≤ 6.93 g
□ **Strip B** B1…B22	22	**212.4 mm** cut 212.7, sand to 212.4 ±0.3	level cut, ~7° off square	level cut, ~7° off square	CG1 ch 1	SJ1	6–11	≤ 6.93 g
□ **Base ring** BASE1…BASE22	22	**14.6 mm** ±0.3 (end gap ≤ 0.6)	lap end, trimmed 5° off square	lap end, trimmed 6° off square	CG1 ch 2	SJ2	1–11	≤ 0.48 g
□ **Ring 1** R1-1…R1-22	22	**13.7 mm** ±0.3 (end gap ≤ 0.6)	lap end, trimmed 6° off square	lap end, trimmed 6° off square	CG1 ch 3	SJ2	12	≤ 0.45 g
□ **Ring 3** R3-1…R3-22	22	**12.9 mm** ±0.3 (end gap ≤ 0.6)	lap end, trimmed 6° off square	lap end, trimmed 7° off square	CG1 ch 4	SJ2	12	≤ 0.42 g
□ **Top cap** CAP1…CAP22	22	**12.8 mm** ±0.3 (end gap ≤ 0.6)	lap end, trimmed 7° off square	lap end, trimmed 7° off square	CG1 ch 5	SJ2	12–13	≤ 0.42 g

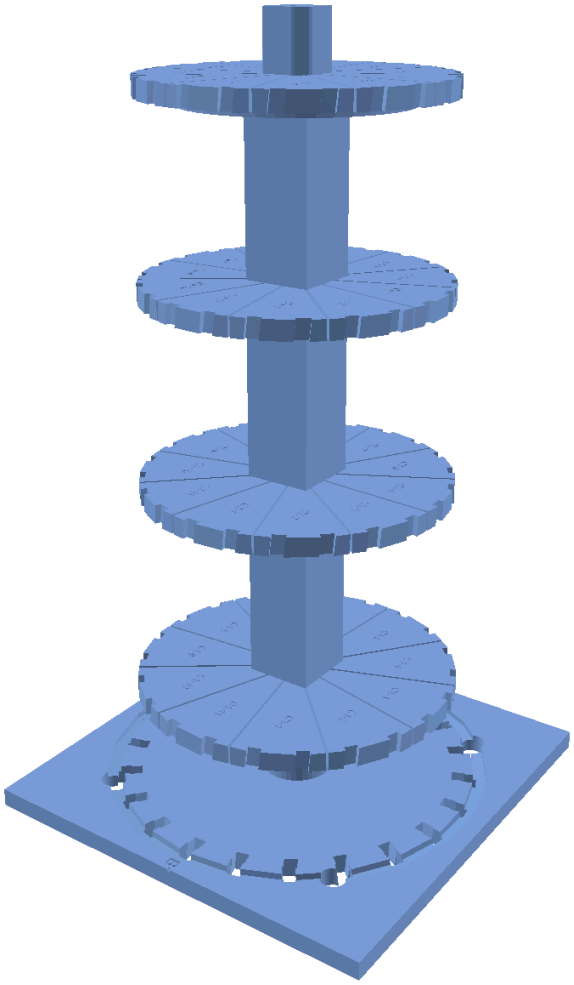
Do this:

1. **Check the fit.** Slide a stick into each slot of FT . Find the smallest slot it slides into with light friction. If that isn't 3.3, set **Jig slot size** in the app to that number and export the build pack again before printing the other jigs.
2. **Sort the sticks.** Weigh each stick and number them 1–13 in pencil, heaviest first. **Go/no-go:** the 13 sticks together should weigh no more than **17.9 g**. Heavier than that and the finished tower will be over 20 g: do the wood test and re-run the design in the app with your density. The stick plan in section 9 puts the strips on sticks 1–11.
3. **Label 6 cups:** Strip A (A1–A22, 22); Strip B (B1–B22, 22); Base ring (BASE1–BASE22, 22); Ring 1 (R1-1–R1-22, 22); Ring 3 (R3-1–R3-22, 22); Top cap (CAP1–CAP22, 22). Every piece goes straight into its cup.
4. **Cut to length** in CG1 (full method in section 9). Square up the stick's end at the slot first. Then for each piece: push the stick along until its end touches the stop, and saw at the slot. No measuring: the stop sets the length.
5. **Letter the strips as they come off the guide.** Each strip lies in the channel with its foot at the left (the slot end).
   - Strip A: write **A** and its number on the **top** face near the left end, then **F** at that end.
   - Strip B: **roll it over** (keep the left end on the left), then write **B** and its number on the new top face near the left end, and **F** at that end.

   The lettered face is the **letter face**: it faces outwards on the tower.
6. **Mark the crossings** in MG1 . Lay each strip in its channel (A or B), letter face up, foot against the angled stop. Pencil a tick through every notch onto the letter face and carry it round onto both side faces. Copy the label beside the notch (1R, 2, 3R; R means a ring goes there). The strip's top end must land between the two small ticks either side of END (±0.3 mm).
7. **Sand the strip ends in SJ1 .** Put the jig on the sandpaper board, letter face of the strip towards the side marked LETTER FACE THIS SIDE. Push the foot (F) end down the slot labelled with its letter and FOOT (for example A FOOT) until it touches the sandpaper. Pinch the strip to the jig so it can't slide, and rub in small strokes. Stop as soon as the plastic all round the slot shows sanding marks: then the end is flush and exact. Do the top end the same way in the matching TOP slot. Check: the strip stands on the table at its lean without rocking. Never push the strip further down while sanding; that only makes it shorter.
8. **Clean up the ring ends in SJ2 .** Each ring end is already cut at its angle by the cutting guide. Find the end's angle in the table (degrees off square) and push the end down the slot with that number that leans the same way as the cut. Hold it still and rub until the plastic scuffs. Only do this if the end is rough or a piece is too long to sit between its strips: a few strokes take off about 0.1 mm.
9. **Count and check.** Count each cup against the table (cut 2–3 spare ring pieces of each size). Weigh each cup: it should be no more than the group mass shown, or swap in pieces from lighter sticks.

⚖ All pieces together should weigh no more than **15.62 g** before any glue.

Step 1. Set up the jig



3D model of this step: `steps/step_01_jig.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:

BP `base_plate.stl`
M `mast.stl`
SP1 `spacer_1.stl`
SP2 `spacer_2.stl`
SP3 `spacer_3.stl`
SP4 `spacer_4.stl`
C1-1 … C1-11 `collar_1_seg1.stl` … `collar_1_seg11.stl` (11 pieces)
C2-1 … C2-11 `collar_2_seg1.stl` … `collar_2_seg11.stl` (11 pieces)
C3-1 … C3-11 `collar_3_seg1.stl` … `collar_3_seg11.stl` (11 pieces)
C4-1 … C4-11 `collar_4_seg1.stl` … `collar_4_seg11.stl` (11 pieces)

Pieces added: none

Do this:

1. Lay a sheet of baking paper on a flat bench and tape the base plate **BP** down on it, pockets facing up.
2. Push the mast **M** into the socket in the middle of the boss. One corner of the mast is cut off, and so is one corner of the socket: line them up and it goes in only one way. Push it down until it stops. If it wobbles, wrap one layer of tape round its foot and push it in again.
3. Build the stack on the mast from the bottom up, using the stack table in section 8: **SP1**, then the segments of collar **C1**, then **SP2**, then **C2**, and so on up to **C4**. Each spacer has the same cut corner in its hole, so it only goes on one way, with its disc on top.
4. Before you cover a spacer's disc, read the numbers printed on it: each number is where that collar segment goes. Put segment **C1-1** on the **1**, **C1-2** on the **2**, and so on. Engraved ID on top, notches facing outwards, inner end touching the mast. The numbers run anticlockwise (seen from above) from the **A1 B1** mark on the base plate.
5. Put packing tape over the base-plate pockets and the base-ring groove, so glue can't bond the tower to the plate. (No glue goes near the collars, so they don't need tape.)
6. Cut 44 tabs of masking tape, about 6 mm × 30 mm, and stick them along the edge of the bench ready for Step 2.

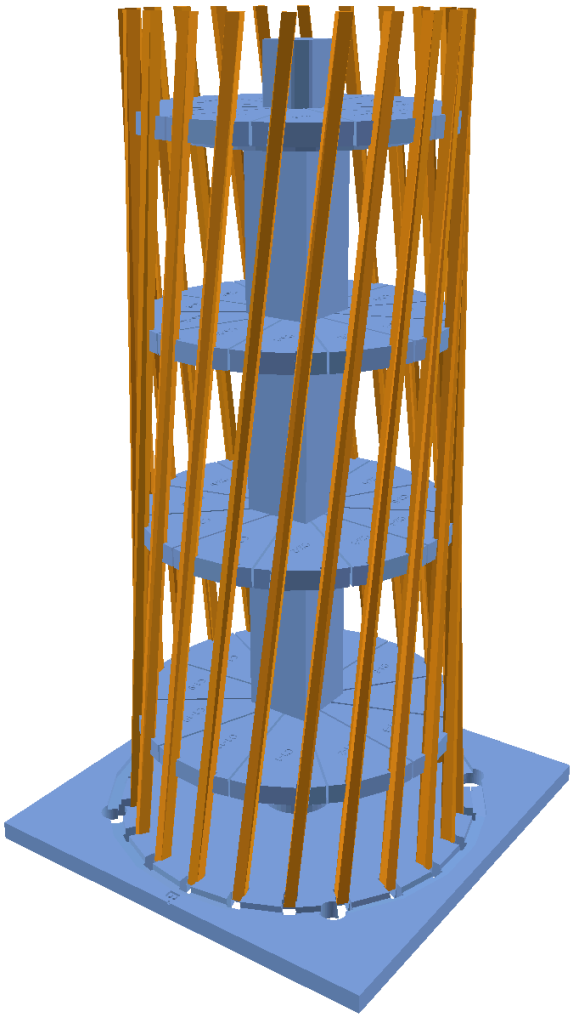
Check:

- □ The mast is upright: sight it against a door frame from two sides.
- □ Every collar is complete and flat, and each segment sits on its own number.
- □ Every segment's ID is on top, and the IDs read C1-…, C2-… from the bottom up.

Common mistakes:

- Stacking the spacers in the wrong order puts the collars at the wrong heights, so the strips won't follow their notches. The disc on each spacer says which collar goes on it ("under C3").

Step 2. Inner strips (set A)



3D model of this step: `steps/step_02_strips_a.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

Pieces added: 22 pieces: A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, A16, A17, A18, A19, A20, A21, A22

Do this:

1. Take the A strips (A1–A22). Each has a letter **A** on its outer face and a foot end marked **F** (Step 0).
2. Seen from outside, the A strips lean to the **right** as they go up (the B strips lean the other way).
3. Each pocket in the base plate is a V: two strips share each foot. **A1** goes in the pocket at the **A1 B1** mark, **A2** in the next pocket to the right (anticlockwise seen from above), and so on round.
4. Stand the strip with its foot in the **inner** arm of the V, letter face outwards, lean it to the right, and press it into the notch of every collar it passes, from C1 up to C4.
5. The tower's surface curves, so a strip has to **twist about 33°** along its length to lie flat in every notch (the notches are angled to match). Wipe each strip with a just-damp cloth first: damp balsa takes the twist without strain. Let the strips dry in the jig for at least 2 hours before any gluing, so they keep the twist.
6. Hold it with a masking-tape tab at collars C2, C3: tape over the strip with the ends pressed onto the top and bottom of the collar. The tabs sit between the B strips' paths, so the B strips never lie on them.
7. Check the angled foot sits flat in its pocket and the top end is 211 mm above the table (measure one).
8. No glue yet. This step only sets out the inner layer.

Check:

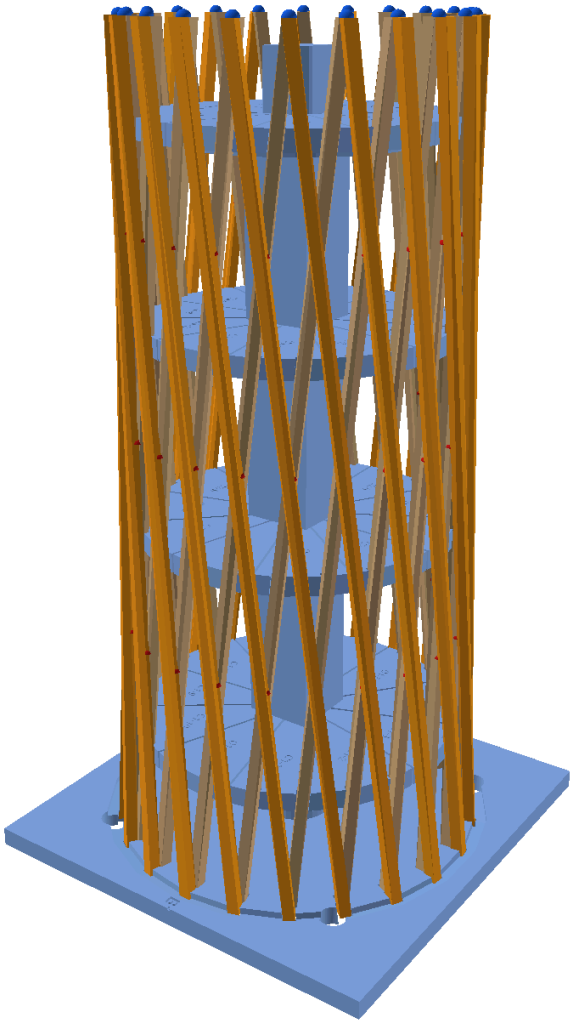
- □ All 22 A strips are in pockets and notches, letter faces outwards.
- □ The strips spiral evenly: the gaps between neighbours look the same all the way up.
- □ The pencil ticks on every strip are at the same heights all round.

Common mistakes:

- A strip in the wrong pocket makes every crossing above it miss. Count the pockets from the mark.

⚖ Weigh these pieces before going on: together they should be no more than **6.93 g**. If they are heavier, swap in lighter sticks now.

Step 3. Outer strips (set B) and the crossings



3D model of this step: `steps/step_03_strips_b.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

Pieces added: 22 pieces: B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12, B13, B14, B15, B16, B17, B18, B19, B20, B21, B22

Glue dots: 🔵 **PVA wood glue** 🔴 **Balsa cement**

Glue in this step:

- 88 × lap with **Balsa cement**: apply ≈ 74 mg each, dries to ≤ 22 mg
- 22 × lap with **PVA wood glue**: apply ≈ 73 mg each, dries to ≤ 37 mg

Do this:

1. Take the B strips (B1–B22). They lean to the **left** going up, seen from outside, and sit in the **outer** arm of each V pocket, letter face outwards. B1 shares A1's pocket.
2. **Dry fit first.** Stand B1 in its pocket and lay it over the A strips. Each pencil tick on B1 should sit right over the crossing with an A strip (the A strip's tick at the same height). Take it out again.
3. **Glue in batches of three strips.** Balsa cement needs two coats. Brush a thin coat on each B strip's **inner** face (opposite the letter) at its 3 ticks and at its foot and top, and on the A strips' outer face at the 5 places each B strip will touch. Wait 5–10 min until it's tacky.
4. Add a second thin coat on the A side, stand the B strip in its pocket and lay it on. Press each crossing for 30 s, working from the bottom up, and **clip a small spring peg across every crossing** (one jaw inside the A strip, one outside the B strip). A straight strip can't sit exactly on every curved crossing: the peg bends it the last 0.1–0.3 mm so the faces close flat. The B strip also gets glued to the A strip at the foot (they share the pocket) and at the top.
5. Like the A strips, the B strips twist slightly to sit flat; wipe them with a damp cloth first.
6. Go round to the right in order: B1–B3, then B4–B6, and so on. That's 110 joints, about 2 hours. Split it over two sessions if you like; stop after a complete batch.
7. Wipe off any glue that squeezes out. It adds mass and no strength.

Check:

- □ Every crossing has glue and a peg, and the faces touch flat (no light between them).
- □ Once pegged, no gap over 0.1 mm at any crossing.
- □ All the masking-tape tabs are still on the outside of the A strips and free of glue.

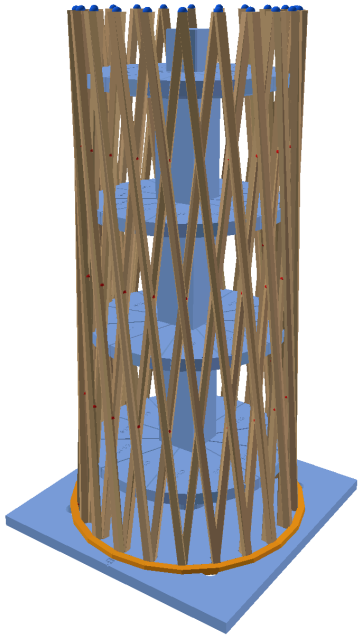
Common mistakes:

- Gluing a crossing before the strip is fully seated in its pocket locks in a crooked strip.
- Coating only one face with balsa cement: it soaks in and the joint stays weak.

⏸ **Wait at least 1 hour** before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **16.61 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 4. Base ring



3D model of this step: `steps/step_04_base.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

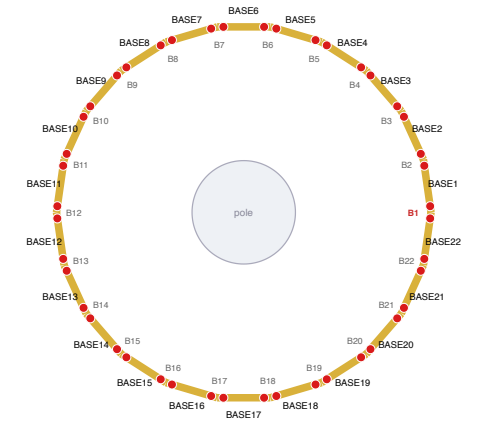
Pieces added: 22 pieces: BASE1, BASE2, BASE3, BASE4, BASE5, BASE6, BASE7, BASE8, BASE9, BASE10, BASE11, BASE12, BASE13, BASE14, BASE15, BASE16, BASE17, BASE18, BASE19, BASE20, BASE21, BASE22

Glue dots: ● **Balsa cement**

Glue in this step:

- 44 × lap end with **Balsa cement**: apply ≈ 12 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; B1 (red) is at the base-plate mark. Dc

Every joint in this ring (44 × Balsa cement):

Piece	End 1 (on strip)	End 2 (on strip)
BASE1	on **B1**: ● Balsa cement	on **B2**: ● Balsa cement
BASE2	on **B2**: ● Balsa cement	on **B3**: ● Balsa cement
BASE3	on **B3**: ● Balsa cement	on **B4**: ● Balsa cement
BASE4	on **B4**: ● Balsa cement	on **B5**: ● Balsa cement
BASE5	on **B5**: ● Balsa cement	on **B6**: ● Balsa cement
BASE6	on **B6**: ● Balsa cement	on **B7**: ● Balsa cement
BASE7	on **B7**: ● Balsa cement	on **B8**: ● Balsa cement
BASE8	on **B8**: ● Balsa cement	on **B9**: ● Balsa cement
BASE9	on **B9**: ● Balsa cement	on **B10**: ● Balsa cement
BASE10	on **B10**: ● Balsa cement	on **B11**: ● Balsa cement
BASE11	on **B11**: ● Balsa cement	on **B12**: ● Balsa cement
BASE12	on **B12**: ● Balsa cement	on **B13**: ● Balsa cement
BASE13	on **B13**: ● Balsa cement	on **B14**: ● Balsa cement
BASE14	on **B14**: ● Balsa cement	on **B15**: ● Balsa cement
BASE15	on **B15**: ● Balsa cement	on **B16**: ● Balsa cement
BASE16	on **B16**: ● Balsa cement	on **B17**: ● Balsa cement
BASE17	on **B17**: ● Balsa cement	on **B18**: ● Balsa cement
BASE18	on **B18**: ● Balsa cement	on **B19**: ● Balsa cement
BASE19	on **B19**: ● Balsa cement	on **B20**: ● Balsa cement
BASE20	on **B20**: ● Balsa cement	on **B21**: ● Balsa cement
BASE21	on **B21**: ● Balsa cement	on **B22**: ● Balsa cement
BASE22	on **B22**: ● Balsa cement	on **B1**: ● Balsa cement

Do this:

1. Take the base-ring pieces (BASE1–BASE22). Each one runs from one foot to the next.
2. Lay a piece on the outside of the B strips, in the shallow groove in the base plate. The strips hold it: the groove leaves 0.25 mm of air all round, so it never touches or sticks to the plastic.
3. Neighbouring pieces meet end to end on a B strip, each covering half its width. Every piece has the arrow you drew when cutting it: **arrow on the outside, pointing right** (its two ends are cut at different angles, so a piece the wrong way round won't sit flush). Glue each end on its strip (dot colour = glue, see the picture). Start at the **A1 B1** mark.

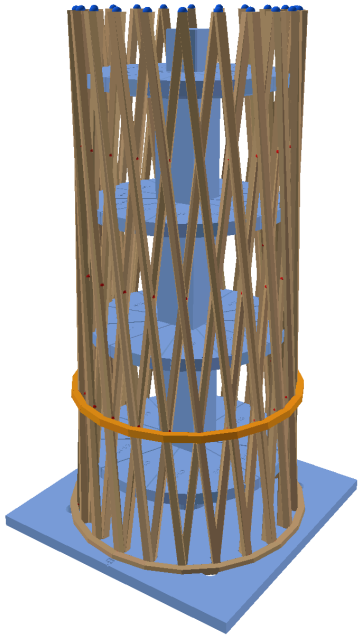
Check:

- □ The base ring sits in its groove all the way round; every piece touches both strips.

⏸ **Wait at least 1 hour** before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **17.25 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 5. Ring 1: 60 mm up



3D model of this step: `steps/step_05_ring_1.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

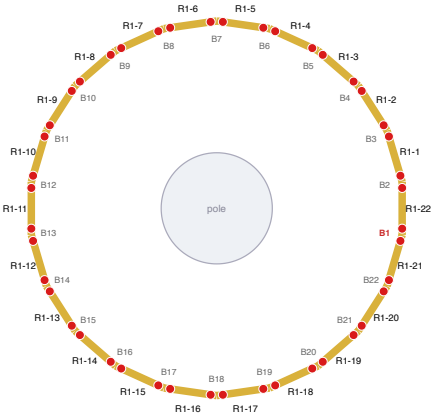
Pieces added: 22 pieces: R1-1, R1-2, R1-3, R1-4, R1-5, R1-6, R1-7, R1-8, R1-9, R1-10, R1-11, R1-12, R1-13, R1-14, R1-15, R1-16, R1-17, R1-18, R1-19, R1-20, R1-21, R1-22

Glue dots: ● **Balsa cement**

Glue in this step:

- 44 × lap end with **Balsa cement**: apply ≈ 12 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; B1 (red) is at the base-plate mark. Dc

Every joint in this ring (44 × Balsa cement):

Piece	End 1 (on strip)	End 2 (on strip)
R1-1	on B2: ● Balsa cement	on B3: ● Balsa cement
R1-2	on B3: ● Balsa cement	on B4: ● Balsa cement
R1-3	on B4: ● Balsa cement	on B5: ● Balsa cement
R1-4	on B5: ● Balsa cement	on B6: ● Balsa cement
R1-5	on B6: ● Balsa cement	on B7: ● Balsa cement
R1-6	on B7: ● Balsa cement	on B8: ● Balsa cement
R1-7	on B8: ● Balsa cement	on B9: ● Balsa cement
R1-8	on B9: ● Balsa cement	on B10: ● Balsa cement
R1-9	on B10: ● Balsa cement	on B11: ● Balsa cement
R1-10	on B11: ● Balsa cement	on B12: ● Balsa cement
R1-11	on B12: ● Balsa cement	on B13: ● Balsa cement
R1-12	on B13: ● Balsa cement	on B14: ● Balsa cement
R1-13	on B14: ● Balsa cement	on B15: ● Balsa cement
R1-14	on B15: ● Balsa cement	on B16: ● Balsa cement
R1-15	on B16: ● Balsa cement	on B17: ● Balsa cement
R1-16	on B17: ● Balsa cement	on B18: ● Balsa cement
R1-17	on B18: ● Balsa cement	on B19: ● Balsa cement
R1-18	on B19: ● Balsa cement	on B20: ● Balsa cement
R1-19	on B20: ● Balsa cement	on B21: ● Balsa cement
R1-20	on B21: ● Balsa cement	on B22: ● Balsa cement
R1-21	on B22: ● Balsa cement	on B1: ● Balsa cement
R1-22	on B1: ● Balsa cement	on B2: ● Balsa cement

Do this:

1. Take the Ring 1 pieces (R1-1–R1-22). This ring goes at crossing level 1, **60 mm** above the table, where the strips' **1R** pencil ticks are.
2. Each piece runs from one crossing to the next, lying on the outside of the B strips. Neighbouring pieces meet end to end on a B strip, each covering half its width. Every piece has the arrow you drew when cutting it: **arrow on the outside, pointing right** (its two ends are cut at different angles, so a piece the wrong way round won't sit flush).
3. Each end crosses its strip right at the tick. You don't need to measure heights: the crossings set them.
4. Start at the strips above the **A1 B1** mark and go round to the right using the glue map and table on this page.

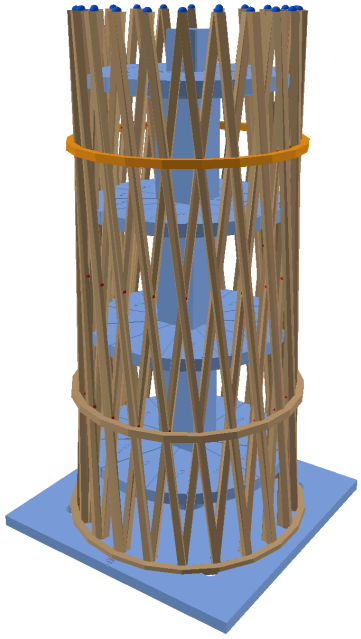
Check:

- ☐ The ring is level; every piece sits flat on the crossings.

⏸ **Wait at least 1 hour** before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **17.86 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 6. Ring 3: 163 mm up



3D model of this step: `steps/step_06_ring_3.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

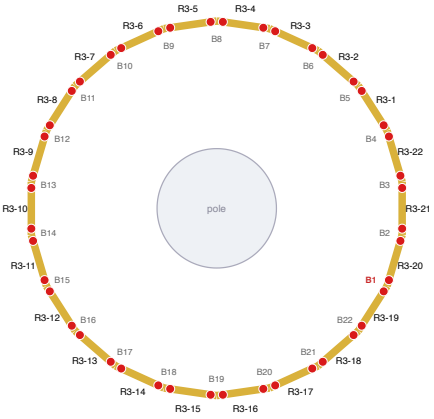
Pieces added: 22 pieces: R3-1, R3-2, R3-3, R3-4, R3-5, R3-6, R3-7, R3-8, R3-9, R3-10, R3-11, R3-12, R3-13, R3-14, R3-15, R3-16, R3-17, R3-18, R3-19, R3-20, R3-21, R3-22

Glue dots: ● **Balsa cement**

Glue in this step:

- 44 × lap end with **Balsa cement**: apply ≈ 12 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; B1 (red) is at the base-plate mark. Dc

Every joint in this ring (44 × Balsa cement):

Piece	End 1 (on strip)	End 2 (on strip)
R3-1	on B4: ● Balsa cement	on B5: ● Balsa cement
R3-2	on B5: ● Balsa cement	on B6: ● Balsa cement
R3-3	on B6: ● Balsa cement	on B7: ● Balsa cement
R3-4	on B7: ● Balsa cement	on B8: ● Balsa cement
R3-5	on B8: ● Balsa cement	on B9: ● Balsa cement
R3-6	on B9: ● Balsa cement	on B10: ● Balsa cement
R3-7	on B10: ● Balsa cement	on B11: ● Balsa cement
R3-8	on B11: ● Balsa cement	on B12: ● Balsa cement
R3-9	on B12: ● Balsa cement	on B13: ● Balsa cement
R3-10	on B13: ● Balsa cement	on B14: ● Balsa cement
R3-11	on B14: ● Balsa cement	on B15: ● Balsa cement
R3-12	on B15: ● Balsa cement	on B16: ● Balsa cement
R3-13	on B16: ● Balsa cement	on B17: ● Balsa cement
R3-14	on B17: ● Balsa cement	on B18: ● Balsa cement
R3-15	on B18: ● Balsa cement	on B19: ● Balsa cement
R3-16	on B19: ● Balsa cement	on B20: ● Balsa cement
R3-17	on B20: ● Balsa cement	on B21: ● Balsa cement
R3-18	on B21: ● Balsa cement	on B22: ● Balsa cement
R3-19	on B22: ● Balsa cement	on B1: ● Balsa cement
R3-20	on B1: ● Balsa cement	on B2: ● Balsa cement
R3-21	on B2: ● Balsa cement	on B3: ● Balsa cement
R3-22	on B3: ● Balsa cement	on B4: ● Balsa cement

Do this:

1. Take the Ring 3 pieces (R3-1–R3-22). This ring goes at crossing level 3, **163 mm** above the table, where the strips' **3R** pencil ticks are.
2. Each piece runs from one crossing to the next, lying on the outside of the B strips. Neighbouring pieces meet end to end on a B strip, each covering half its width. Every piece has the arrow you drew when cutting it: **arrow on the outside, pointing right** (its two ends are cut at different angles, so a piece the wrong way round won't sit flush).
3. Each end crosses its strip right at the tick. You don't need to measure heights: the crossings set them.
4. Start at the strips above the **A1 B1** mark and go round to the right using the glue map and table on this page.

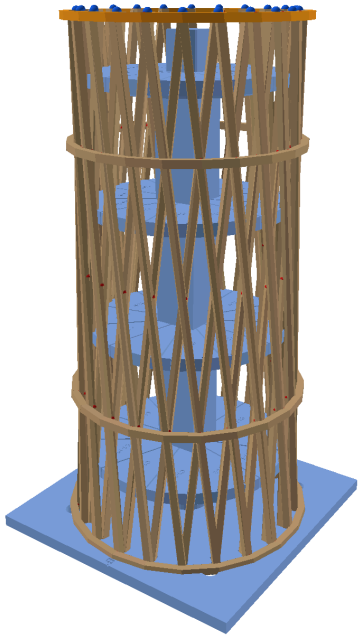
Check:

- ☐ The ring is level; every piece sits flat on the crossings.

⏸ **Wait at least 1 hour** before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **18.45 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 7. Top ring



3D model of this step: `steps/step_07_cap.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

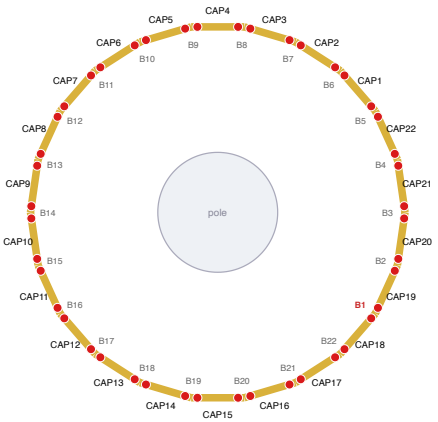
Pieces added: 22 pieces: CAP1, CAP2, CAP3, CAP4, CAP5, CAP6, CAP7, CAP8, CAP9, CAP10, CAP11, CAP12, CAP13, CAP14, CAP15, CAP16, CAP17, CAP18, CAP19, CAP20, CAP21, CAP22

Glue dots: ● **Balsa cement**

Glue in this step:

- 44 × lap end with **Balsa cement**: apply ≈ 12 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; B1 (red) is at the base-plate mark. Dc

Every joint in this ring (44 × Balsa cement):

Piece	End 1 (on strip)	End 2 (on strip)
CAP1	on B5: ● Balsa cement	on B6: ● Balsa cement
CAP2	on B6: ● Balsa cement	on B7: ● Balsa cement
CAP3	on B7: ● Balsa cement	on B8: ● Balsa cement
CAP4	on B8: ● Balsa cement	on B9: ● Balsa cement
CAP5	on B9: ● Balsa cement	on B10: ● Balsa cement
CAP6	on B10: ● Balsa cement	on B11: ● Balsa cement
CAP7	on B11: ● Balsa cement	on B12: ● Balsa cement
CAP8	on B12: ● Balsa cement	on B13: ● Balsa cement
CAP9	on B13: ● Balsa cement	on B14: ● Balsa cement
CAP10	on B14: ● Balsa cement	on B15: ● Balsa cement
CAP11	on B15: ● Balsa cement	on B16: ● Balsa cement
CAP12	on B16: ● Balsa cement	on B17: ● Balsa cement
CAP13	on B17: ● Balsa cement	on B18: ● Balsa cement
CAP14	on B18: ● Balsa cement	on B19: ● Balsa cement
CAP15	on B19: ● Balsa cement	on B20: ● Balsa cement
CAP16	on B20: ● Balsa cement	on B21: ● Balsa cement
CAP17	on B21: ● Balsa cement	on B22: ● Balsa cement
CAP18	on B22: ● Balsa cement	on B1: ● Balsa cement
CAP19	on B1: ● Balsa cement	on B2: ● Balsa cement
CAP20	on B2: ● Balsa cement	on B3: ● Balsa cement
CAP21	on B3: ● Balsa cement	on B4: ● Balsa cement
CAP22	on B4: ● Balsa cement	on B5: ● Balsa cement

Do this:

1. Take the top-ring pieces (CAP1–CAP22).
2. Lay each piece on the outside of the B strips, with its top flush with the strip tops. Neighbouring pieces meet end to end on a B strip, each covering half its width. Every piece has the arrow you drew when cutting it: **arrow on the outside, pointing right** (its two ends are cut at different angles, so a piece the wrong way round won't sit flush). Glue both ends (dot colour = glue).
3. Start at the strips above the **A1 B1** mark and go round to the right, using the glue map and table on this page.

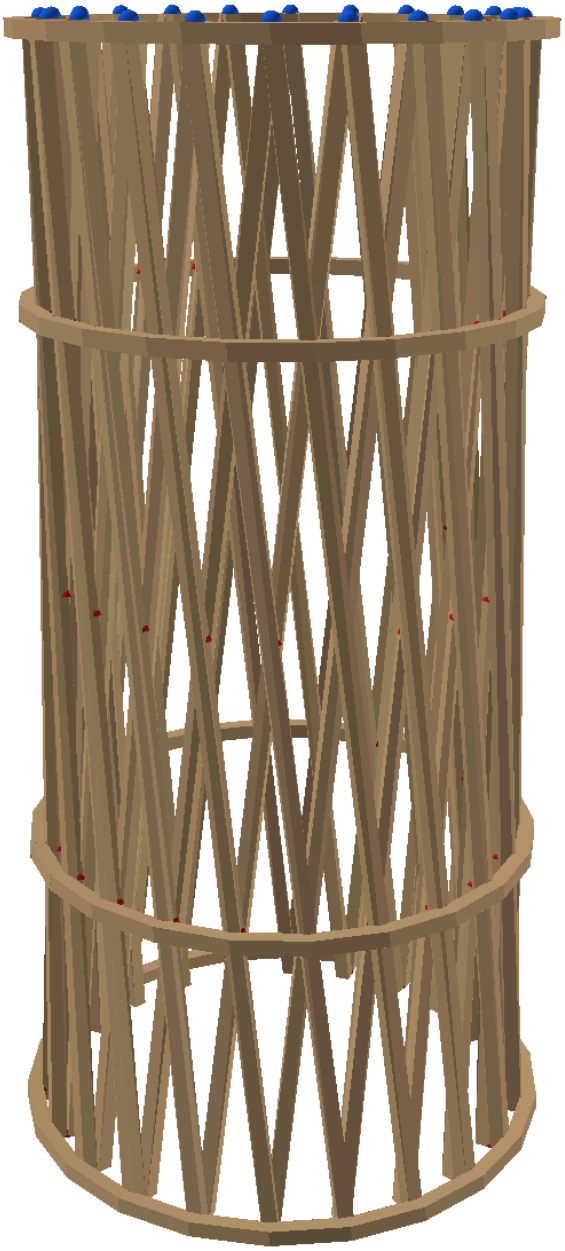
Check:

- ☐ The top ring is flush with the strip ends all the way round.

⏸ **Wait at least 1 hour** before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **19.03 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 8. Cure, remove the jig and finish



Jig files on the bench:
none (the jig comes off in this step)

Pieces added: none

3D model of this step: `steps/step_08_remove.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Do this:

1. Leave everything for **24 hours** so every glue reaches full strength.
2. Peel off every masking-tape tab.
3. Lift the tower off the base plate, mast and all: push the feet up gently through the finger holes, a little at a time all round, until they come out of the pockets. Hold the tower by its rings, never by one strip.
4. Hold the tower over a box and pull the mast **M** straight up out of the top.
5. Now take the jig out **from the bottom up**, in the reverse of how you stacked it: slide **SP1** down and out; then push each **C1** segment about 3 mm inwards off its strips, tip it and lower it out of the bottom; then **SP2**, then the **C2** segments, and so on up to **C4**. Every part comes out through the wide bottom of the tower.
6. Never pull a segment outwards or upwards through the strips. If one sticks, push it inwards firmly or cut it with a knife: it's only a jig part.
7. Check the tower on a flat table. If it rocks, lay the sandpaper board on the table, rest a flat book on the top ring, and slide the tower in short straight strokes (2–3 cm, no twisting) with light pressure. Check after every 5 strokes.
8. Do the same for the top, upside down, so the load plate will bear on every strip at once.
9. Weigh the finished tower.

Check:

- □ It weighs no more than the limit.

- □ It stands upright without rocking.
- □ The pole slides through without touching.

Common mistakes:

- Sanding in circles or pressing hard snaps feet. Short straight strokes, light pressure.

11. Final checks

- □ Mass is no more than **19.03 g** (limit 20 g).
- □ It stands on a flat table without rocking; every foot touches.
- □ The top is flat: lay a ruler across it in three directions and check for gaps.
- □ No joint has a visible gap or loose piece. Press gently on each ring; nothing moves.
- □ The pole slides through without touching (about 24 mm gap).
- □ No masking tape or jig parts left inside.

12. Test day

1. Weigh the tower in front of the tester and record it.
2. Slide it over the pole, centred. Check the gap all round.
3. Seat the load plate gently and check it touches the top ring and strip tops all the way round. Don't shim; sand if needed.
4. Add load smoothly and evenly. Shock loads break brittle joints first.
5. Record the load at failure and where it failed (which piece or joint). Compare it with the prediction (60 kg) and the weakest point listed above.

Tower mass (g)	Failure load (kg)	Where it failed	Efficiency (load ÷ mass)

13. Troubleshooting

Problem	Likely cause	Fix
Strips won't sit in the collar notches	A segment on the wrong number, upside down, or spacers swapped	Check each segment sits on its own number with its ID on top, and each spacer's disc says "under" the collar above it.
A crossing doesn't touch (gap)	Strip not fully seated, or bowed	Re-seat the strip in its notches. Swap bowed strips for straight ones. Never fill a gap with glue.
Over mass at a weigh-in	Too much glue	Use less on the remaining joints; wipe squeeze-out immediately. Do the glue test.
Glue stuck the tower to the jig	Glue ran onto the plastic	Free it with a thin blade; sand the joint smooth. Keep packing tape on the base plate.
Ring not level	Pieces glued at slightly different heights	Check each end sits on its pencil tick before the glue sets.
Collar segment won't come out	Glue on the notch	Push it inwards firmly; if stuck, cut the segment with a knife. It is only a jig part.
Tower rocks on the table	Feet not level	Short straight strokes on the sandpaper board with a book on top (Step 8). Never circles.
Balsa cement joint stays soft	Single coat, or not cured	Use the double-glue method and wait 12 h.

Fixing mistakes

What went wrong	What to do
A strip breaks before it's glued	Cut a new one from a spare stick in channel 1, letter it, mark it in the marking gauge and sand its ends, exactly like the others.
A glued piece breaks, or a joint is glued in the wrong place	Balsa cement softens with acetone: dab a little on the joint with a cotton bud, wait a minute, and ease the pieces apart. Scrape off the old glue, let it dry, and glue again. Replace a broken piece with a spare.
A ring piece too short (more than 0.6 mm gap at its ends)	Use one of the spare ring pieces.
A strip end sanded too far (strip short of the END notch)	More than 0.3 mm short (outside the small ticks beside END): cut a new strip. The tops must be level, and the tower must stay over 210 mm.
You run out of sticks	You have 2 spare; the offcuts at the end of the strip sticks are long enough for ring pieces.
A jig part is lost or broken	Reprint just that file; its name is in the table in section 8 and Appendix B.
Over mass at the end	Don't sand pieces thinner. Look for big glue blobs and scrape them off with a blade.

Appendix A: every piece

ID	Group	Length mm	End 1	End 2	Glue at its joints	Max mg
A1	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A2	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A3	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A4	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A5	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A6	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A7	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A8	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A9	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A10	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A11	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A12	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A13	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A14	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A15	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A16	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A17	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A18	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A19	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A20	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A21	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
A22	Strip A	212.4	level cut, ~7° off square	level cut, ~7° off square	4× Balsa cement, 1× PVA wood glue	315
B1	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B2	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B3	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B4	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B5	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B6	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B7	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B8	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B9	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B10	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B11	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B12	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B13	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B14	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B15	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B16	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B17	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B18	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B19	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B20	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B21	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
B22	Strip B	212.4	level cut, ~7° off square	level cut, ~7° off square	12× Balsa cement, 1× PVA wood glue	315
BASE1	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE2	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE3	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE4	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE5	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE6	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE7	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE8	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22

BASE9	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE10	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE11	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE12	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE13	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE14	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE15	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE16	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE17	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE18	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE19	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE20	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE21	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
BASE22	Base ring	14.6	lap end, trimmed 5° off square	lap end, trimmed 6° off square	2× Balsa cement	22
CAP1	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP2	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP3	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP4	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP5	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP6	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP7	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP8	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP9	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP10	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP11	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP12	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP13	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP14	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP15	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP16	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP17	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP18	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP19	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP20	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP21	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
CAP22	Top cap	12.8	lap end, trimmed 7° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R1-1	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-2	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-3	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-4	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-5	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-6	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-7	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-8	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-9	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-10	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-11	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-12	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-13	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-14	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-15	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-16	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-17	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-18	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-19	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20

R1-20	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-21	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R1-22	Ring 1	13.7	lap end, trimmed 6° off square	lap end, trimmed 6° off square	2× Balsa cement	20
R3-1	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-2	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-3	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-4	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-5	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-6	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-7	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-8	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-9	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-10	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-11	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-12	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-13	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-14	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-15	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-16	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-17	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-18	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-19	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-20	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-21	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19
R3-22	Ring 3	12.9	lap end, trimmed 6° off square	lap end, trimmed 7° off square	2× Balsa cement	19

Appendix B: jig and model files

Jig files (`jig/` ; each has a .stl to print and a .step for Onshape) and the steps that use them:

File	Used in step
`base_plate.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_1_seg11.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_2_seg11.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_3_seg11.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7
`collar_4_seg11.stl / .step`	1, 2, 3, 4, 5, 6, 7
`cutting_guide_1.stl / .step`	Step 0 (preparing parts)
`end_sanding_jig_7deg.stl / .step`	Step 0 (preparing parts)
`end_sanding_jig_rings.stl / .step`	Step 0 (preparing parts)
`fit_test_coupon.stl / .step`	Step 0 (preparing parts)
`marking_gauge.stl / .step`	Step 0 (preparing parts)

`mast.stl / .step`	1, 2, 3, 4, 5, 6, 7
`spacer_1.stl / .step`	1, 2, 3, 4, 5, 6, 7
`spacer_2.stl / .step`	1, 2, 3, 4, 5, 6, 7
`spacer_3.stl / .step`	1, 2, 3, 4, 5, 6, 7
`spacer_4.stl / .step`	1, 2, 3, 4, 5, 6, 7

Step models (`steps/` , open in Onshape):

Step	What	File
Step 1	Set up the jig	`steps/step_01_jig.step`
Step 2	Inner strips (set A)	`steps/step_02_strips_a.step`
Step 3	Outer strips (set B) and the crossings	`steps/step_03_strips_b.step`
Step 4	Base ring	`steps/step_04_base.step`
Step 5	Ring 1: 60 mm up	`steps/step_05_ring_1.step`
Step 6	Ring 3: 163 mm up	`steps/step_06_ring_3.step`
Step 7	Top ring	`steps/step_07_cap.step`
Step 8	Cure, remove the jig and finish	`steps/step_08_remove.step`

Whole tower: `tower.step` . **Assembled jig:** `jig/jig_assembly.step` .

Importing into Onshape: in a document click **+** → **Import…** and choose the file. STEP files come in as proper editable parts; STL files come in as meshes (fine for viewing, awkward to edit).